



Coupling agent-derived tailored Econeas fouling-release coating with superior tolerance

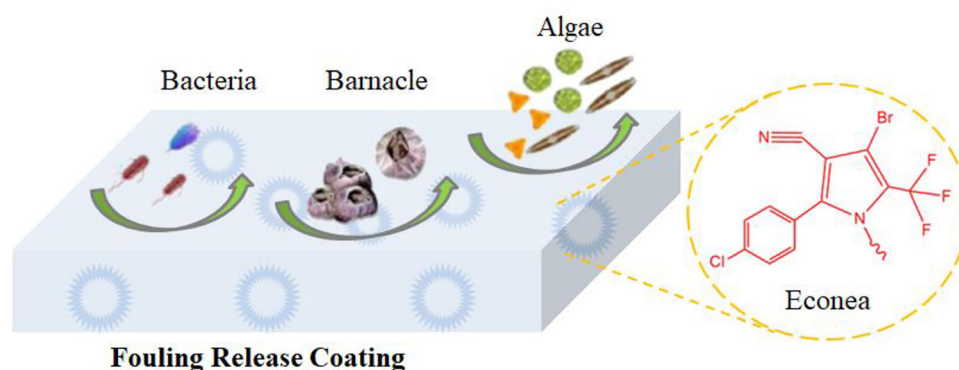
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Abstract The most effective strategy for controlling biofouling is the release of toxic antifouling agents, but this approach triggers environmental concerns and leads to stricter international legislation. This study proposes a novel, simple, and convenient method for grafting antifouling agents by using a silane coupling agent to introduce Econeas into low-surface-energy coatings. The amount of introduced antifouling agent can be precisely controlled. Compared with traditional antifouling agent-releasing coatings, the leaching rate of the grafted antifouling coating is only one-sixth that

of conventional coatings, having maintained satisfactory antifouling performance for 141 days in real-world marine trials. Furthermore, this study demonstrates that introducing Econeas into low-surface-energy coatings via a coupling agent does not significantly alter its original antifouling properties. These findings not only address industry concerns but also establish a framework for the grafting technology of novel antifouling agents.

Graphic abstract



Silane coupling agent graft coating shown is highly resistant to dirt from marine biological organisms.

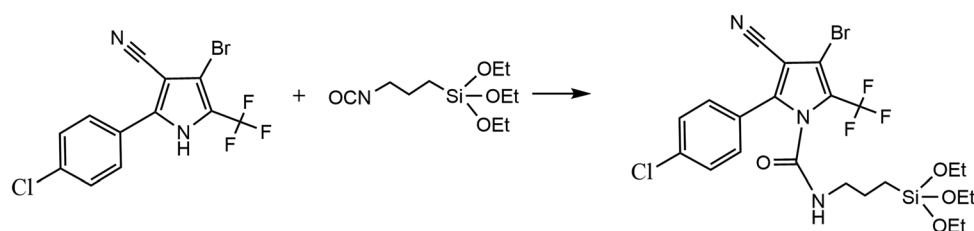
Keywords Antifouling, Foul release, Econeas, Grafting, Chemical immobilization

Introduction

Biofouling poses significant challenges to marine operations and coastal infrastructure, particularly in maritime industries.^{1,2} To address this persistent issue,

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Scheme 1: Synthesis of grafted curing agent

Table 1: Compositions of fouling-release coatings grafted with Econe

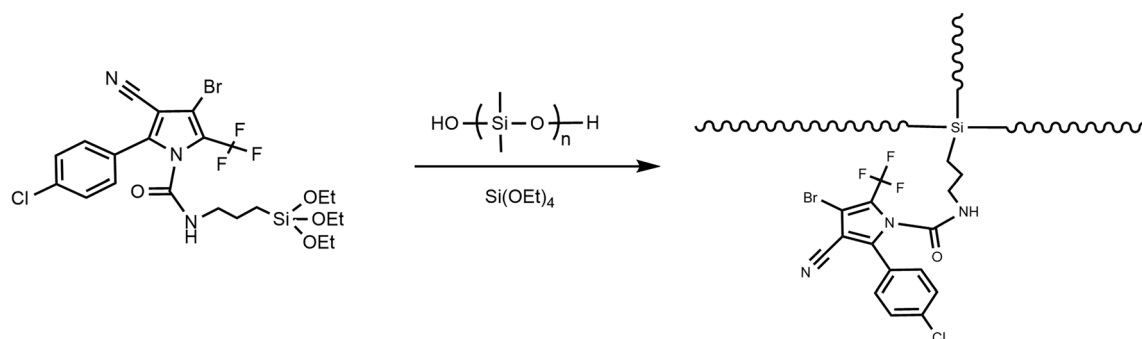
Coating	S0	S1	S2	S3	S4	S4c
821 / g	60.0	60.0	60.0	60.0	60.0	60.0
Econe reagent / g	0	2.9	5.8	8.7	11.6	0
KH550 / g	0	0	0	0	0	1.6
TEOS / g	1.2	1.0	0.8	0.6	0.4	0.4
Econe content /%	0	0.5	1.0	1.5	2.0	2.0

researchers have developed various strategies including surface engineering and biomimetic designs, with emerging technologies demonstrating potential for eco-friendly solutions.^{3–5} While effective in the short term, conventional antifouling methods predominantly rely on chemical control mechanisms—either through direct biocidal coatings or controlled release of toxic compounds like copper oxides and organotin derivatives into adjacent waters.⁶ However, this approach raises growing environmental concerns as continuous leaching of persistent biocides not only reduces treatment longevity but also leads to bioaccumulation in marine ecosystems, ultimately disrupting aquatic food chains and exacerbating environmental toxicity risks.^{7,8}

The pursuit of sustainable antifouling technologies has yielded three principal eco-conscious approaches:^{9–12} (1) non-biocidal foul-release coatings employing low-surface-energy polymers like polydimethylsiloxane (PDMS) to prevent organism adhesion;¹³ (2) controlled-release systems integrating microencapsulated green biocides (e.g., capsaicin derivatives, medetomidine) within hybrid resin matrices for timed elution;¹⁴ (3) bioactive contact inhibition through covalently immobilized antimicrobial peptides or enzyme-mimetic polymers.¹⁵ The first strategy has been well accepted due to its inherently low environmental impact, but it presents some drawbacks, such as high cost, mechanical fragility, and restricted efficacy.¹⁶ These limitations make it less suitable for non-static conditions, such as fast-moving vessels traveling at speeds greater than 15 knots.¹⁷ The second strategy has also been widely adopted. An example is biocidal foul-

release coatings, which combine both antifouling and foul-release properties in a single paint matrix, e.g., Hempguard marine paint (X7, X8) from Hempel can be considered a representative of this approach.¹⁸ It combines conventional foul-release coatings with microencapsulated biocides. Although this paint provides a considerable extension of the coating life cycle (up to 90 months) and reduces biocide release, it is still based on a biocide-release strategy and thus faces current environmental constraints. As is known, X7 and X8 are high-end antifouling paints.¹⁶ There are also many mid-to-low-end antifouling paints that release more biocides, even though these are green antifouling agents. The latter approach has also been of particular interest in the last few years, and some promising results have already been reported.^{19,20} New strategies that combine the benefits of more than one of these emerging approaches have also been proposed.²⁰ The direct incorporation of biocides into polydimethylsiloxane (PDMS)-based foul-release matrices has been attempted, but incompatibility issues were encountered.²¹ Additionally, the strategy of incorporating grafted antifouling agents into silicone resins and combining them with a nontoxic antifouling coating system has been explored.²² However, the high cost and complex process have limited its application.

The persistent challenges of biocidal ecotoxicity and mechanical failure in current antifouling systems necessitate the development of cost-effective, facile-to-fabricate alternatives that combine scalable processing with tunable performance while maintaining ecological compatibility. Aligned with these global aims, this work presents a simple, efficient, and tailored alternative based on the Econe biocide grafted in silicone resin matrix. The developed coating formulations demonstrate superior tolerance and long-term antifouling effects under real-world conditions. Moreover, these non-release coatings significantly reduce the leaching of biocides from the polymeric matrix. These findings provide significant insights into optimizing low-surface-fouling coatings, which not only address existing limitations but also exhibit unexpected inhibitory effects under conditions with low biocide content. This clearly indicates the potential for broader applications of grafting technology in the development of advanced antifouling agents.



Scheme 2: Synthesis of Econe-grafted fouling-release antifouling coating

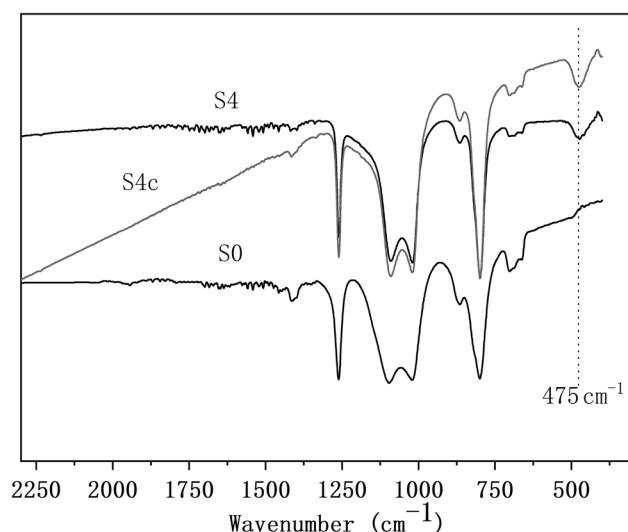


Fig. 1: IR spectra of coatings

Experimental section

Materials

4-Bromo-2-(4-chlorophenyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile (Econe), tetraethoxysilane (TEOS), ethyl acetate, di-n-butyltindilaurate (DBTDL), vinyltriethoxysilane (VTEO), aminopropyltriethoxysilane (KH550), xylene (Xy), and 3-isocyanatopropyltriethoxysilane (IPTS) were received from Shanghai Titan Technology Co., Ltd. (Shanghai, China). 821 Hydrogen silicone oil (821) was received from Quzhou Haina New Material Technology Co., Ltd. (Zhejiang, China). Econe was dried under vacuum at 40 °C for 24 h before use. Xy was refluxed and then distilled over sodium. Ethyl acetate was dehydrated using 4A molecular sieve and then used after distillation. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2021).²³

Preparation of low-surface antifouling coating with grafted Econe

Preparation of silane coupling agents grafted with Econe (Econe reagent)

Under anhydrous argon protection, 25.0 g of dried Econe was completely dissolved in 100.0 g of dried ethyl acetate solution. Then, the solution was added dropwise and uniformly over a period of 6 h to an 18.6 g solution of isocyanate propyltriethoxy silane at 45 °C. After the addition was completed, the reaction was allowed to proceed until the NCO groups were completely consumed (as detected by nitrogen titration), resulting in the silane coupling agent grafted with Econe. The synthetic route is shown in the figure below (Scheme 1).

Preparation of fouling-release coatings grafted with Econe

The following amounts of 821 hydroxyl silicone oil, Econe grafting reagent, ethyl silicate, xylene, and catalyst DBTDL were mixed uniformly and then cured at 45 °C for 24 h (Table 1). After curing, the mixture was dried in a vacuum oven at 45 °C for 48 h to obtain the low-surface-energy antifouling coating grafted with bromopyrrole nitriles. The maximum Econe loading did not exceed 2.0 wt%. When the Econe content reached 2.5 wt%, the system failed to cure completely within 72 h under the given conditions. The synthetic route is shown in Scheme 2.

Swelling ratio of coatings

A gravimetric method was employed to monitor in-situ the swelling of each coating (sample 0, 1, 2, 3, 4 and 4c). Each glass slide was weighed before and after coating to determine the initial dry weight (W_{dry}). Three slides of each coating were used to obtain an average value and deviation. The coated glass slides were immersed in an ASW container which was renewed every other day, and each slide was weighed

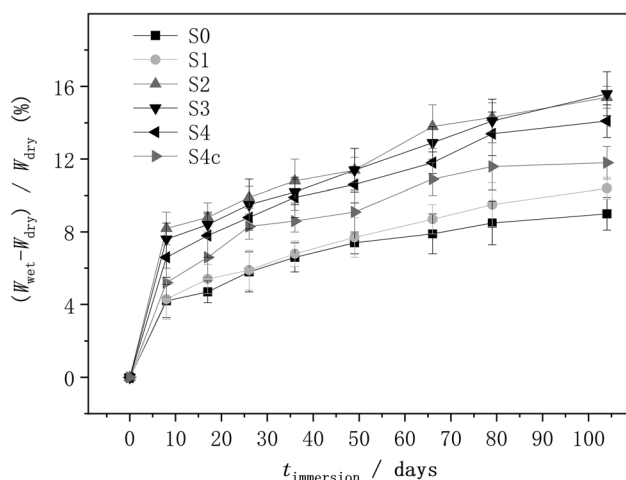


Fig. 2: Swelling ratio of coatings contact angle of coatings

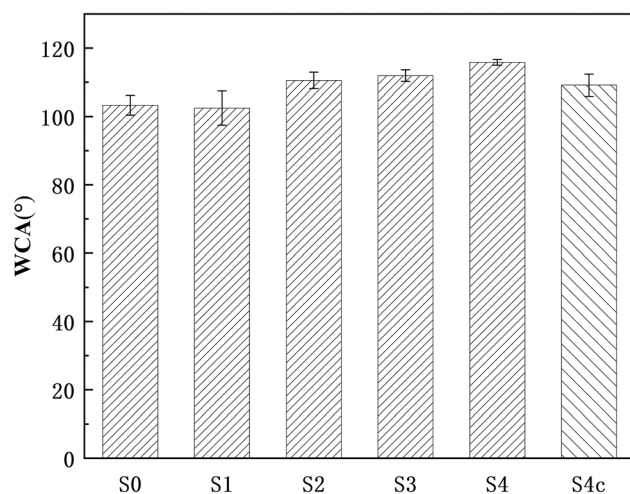


Fig. 3: Contact angle of coatings

after certain periods (W_{wet}). Both W_{dry} and W_{wet} were adjusted by subtracting the initial weight of the glass slide (W_0).

Characterization

IR spectrum was recorded on a Fourier transform infrared (FTIR) spectrometer (TENSOR II, BRUKER, USA). Analyses were performed in the transmission mode in the range $4000\text{--}500\text{ cm}^{-1}$ at room temperature with a resolution of 2 cm^{-1} and accumulation of 16 scans. Surface roughness was assessed by using an ultra-depth-of-field electron microscopy (KEYENCE VHX-6000, Keyence Corporation, Osaka, Japan). Contact angle measurements were performed by using a contact angle measuring instrument (SINDIN SDC-200S, Dongguan Shengding Precision

Instrument Co. LTD., Dongguan, China) with an automatic dispenser. The contact angle of each sample was measured after it was taken out from the sea water, tilted and placed for a period of time (e.g., 30 min here), and dried naturally on the surface. A droplet (about 25 μL) of Milli-Q water was placed on the surface of each sample; then, the liquid was withdrawn immediately. The average of right and left contact angles (denoted “contact angle” in this work) was measured and calculated from three pictures taken each second, using Young–Laplace fitting method. Contact angles of four different spots on the surface were measured, and the final value of contact angle was the average of contact angles at these four spots.

The release rate of Ectonea

The release rate of Ectonea was determined by measuring its concentration using UPLC under a static condition. Typically, a composite coating of antifouling coating was prepared on a glass panel ($25 \times 75\text{ mm}^2$), and then, the panel was immersed in artificial seawater (ASW). After a given period of time, the panel was transferred from the holding tank to an individual measuring container with 100 mL of fresh seawater. After 24 h immersion, 10 mL of seawater was taken from the individual measuring container and extracted three times using 10 mL of trichloromethane. After drying under nitrogen gas, the extract was resuspended in 100 μL of methanol and then subjected to HPLC analysis. The Ectonea peaks were detected from its retention time, and its amount was calculated from the established standard curves using peak areas plotted against known quantities of standards. The recovery efficiency for Ectonea analysis was 83.8%.

Algae inoculation test

Na_2SiO_3 was added to the artificial seawater to achieve a concentration of 7.3 mg/L as the silicon source for diatom growth. The algae were cultured at $25\text{ }^\circ\text{C}$ for 48 h. Coated glass slides were placed in the algal suspension at $25\text{ }^\circ\text{C}$. After 2 days, the biofilm on the coating surface was gently rinsed with artificial seawater to remove unsettled algae. The algal coverage on the coating surface was observed using an inverted fluorescence microscope (XD, Sunny Optical Technology), photographed, and the percentage of coverage area was calculated.

Marine field tests

The antifouling capacity test is carried out in accordance with the national marine antifouling coatings shallow sea experimental standard. Each coating was

applied on the substrate, $300 \times 250 \times 3 \text{ mm}^3$, at the bottom was epoxy zinc rich primer, and the finish coating was our experimental antifouling paint as prepared for testing. Three sets of parallel samples were prepared for each coating to test their antifouling capability at different depths in the ocean. After coating, the four corners of samples were fixed on the hanger with stainless steel bolts. After the installation completed, the hanger was placed on the experimental floating raft in the testing sea area. The experimental sea area is located in the offshore of Louman's sheltered bay, Zhoushan, a subtropical sea; the velocity of sea water is lower than 5 m/s, the azimuth is $30^\circ 01' \text{ N}$, $122^\circ 06' \text{ E}$, the test depth is 0.3–2.0 m, the testing lasted for 141 days, and the main fouling organism attached to the board are algae and silt.

Results and discussion

IR spectra of coating surface

The IR spectra (Fig. 1) of the S0 without Econeal are highly similar to those of S4 and S4c. The small, weak, and broad peak at 1415 cm^{-1} is attributed to the in-plane bending vibration absorption peak of C–H or –OH. The sharp and strong peak at 1260 cm^{-1} corresponds to the symmetric deformation vibration absorption peak of Si–CH₂. The strong and broad peak at 1100 cm^{-1} is due to the Si–O stretching vibration absorption peak. The strong and broad peak at 1020 cm^{-1} is attributed to the linear Si–O–Si stretching vibration absorption peak. The peak at 796 cm^{-1} is due to the stretching vibration absorption peak of the Si–C bond. The peak at 698 cm^{-1} is the characteristic absorption peak of di-substituted benzene. The only

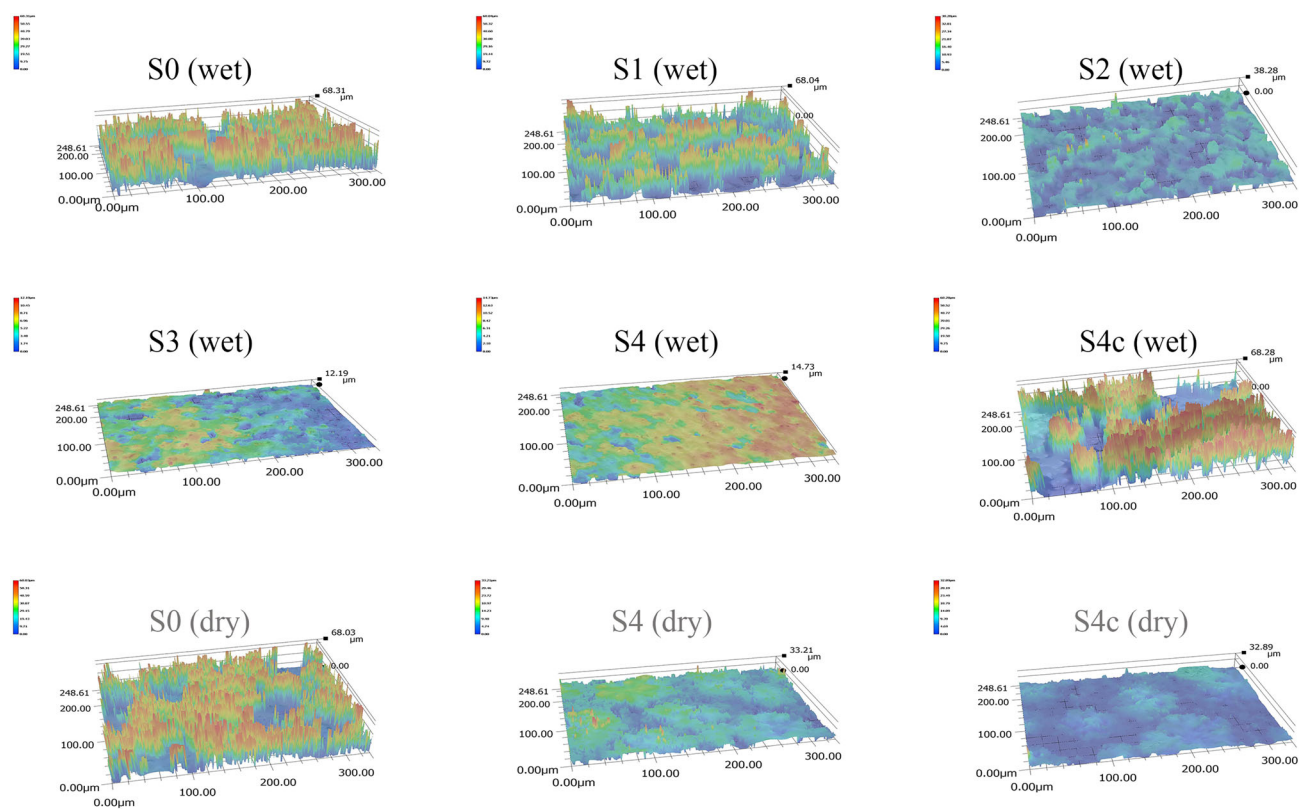


Fig. 4: Extended depth of field (EDF) EM image of coatings

Table 2: Econeal content detected by UPLC-MS (30 days immersion in ASW)

Coating	Econeal content	
	In coating formulation (wt%)	In ASW (μg/L)
S4 (grafted Econeal)	2.00 ± 0.05	0.432 ± 0.103
S4c (blending)	2.00 ± 0.05	2.583 ± 0.742

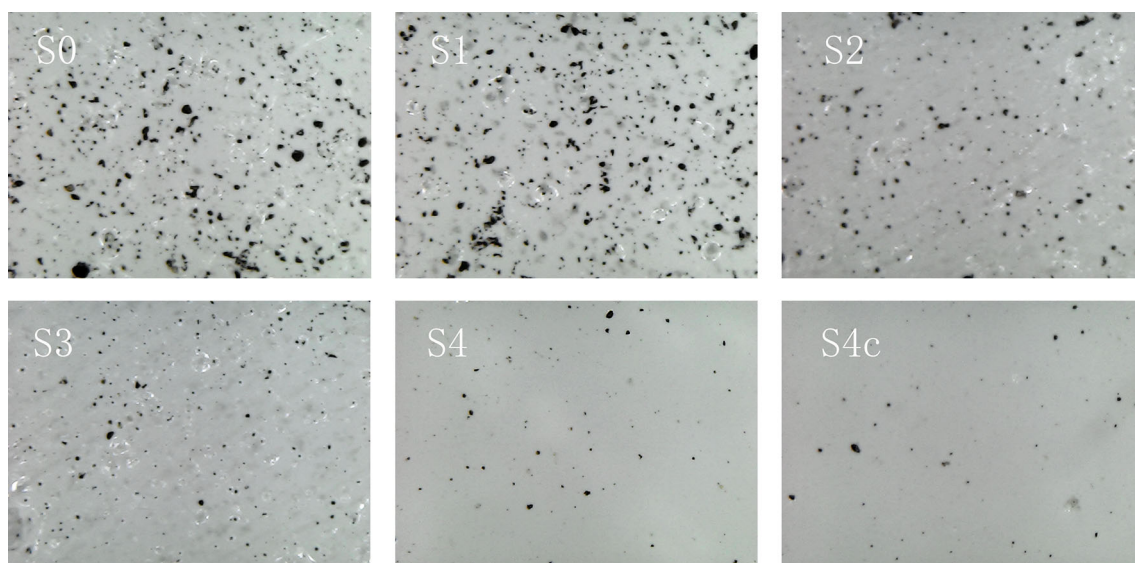


Fig. 5: Antialgae performance of coatings

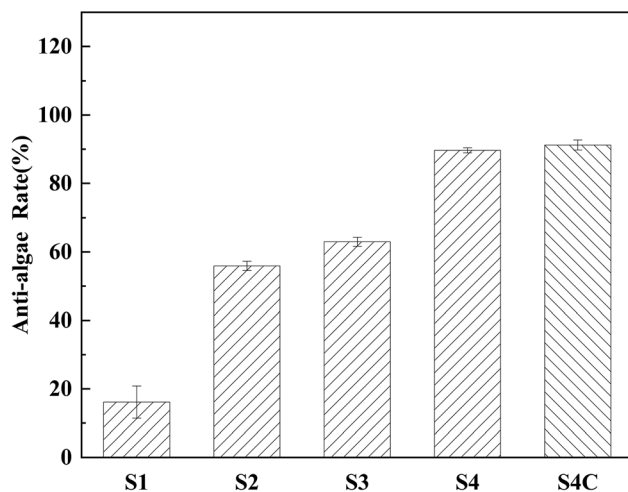


Fig. 6: Antialgae rate of coatings

difference in the IR spectra is that a weak and broad absorption peak appears at 475 cm^{-1} in S4 and S4c, which is not present in S0. This peak may be due to the bending vibration absorption peak of the C-Br bond. Additionally, the absence of the NCO characteristic peak at 2250 cm^{-1} in the IR spectra indicates that NCO has reacted completely.

Swelling ratio of coatings

As can be seen from Fig. 2, the swelling ratio of the samples increases moderately with the increasing amount of E-conea introduced. The possible reason is that the pendant E-conea increases the pore size between the silicone molecules, making it easier for

water to penetrate into the elastomer.²⁴ The swelling ratio of sample S4 is higher than that of S4c, which may be due to the better stability of S4, where E-conea is grafted chemically. In contrast, E-conea in S4c is physically blended, and its inherent hydrophilicity may cause it to partially leach into the water. This is further supported by the subsequent E-conea release test, where the concentration of E-conea in the S4c solution was six times higher than that in the S4 solution, which also demonstrates the stability of the chemically grafted E-conea.

All samples exhibited a hydrophobic contact angle (Fig. 3). In sample S1, where the E-conea grafting amount was 0.5%, the contact angle was almost the same as that of the control sample S0 (without E-conea), both around 102° . When the grafting amount of E-conea was increased to 1%, the contact angle quickly increased from 102 to 110° . As the grafting amount of E-conea continued to increase, the contact angle also continued to increase. When the grafting amount of E-conea was 2%, the contact angle reached 115° . If E-conea is physically blended at 2%, the contact angle can reach 109° . It can be seen that the method of chemical grafting is more effective in increasing the contact angle of the coating, making the coating more hydrophobic.²⁵ The possible reason is that the N-H bonds of E-conea react with NCO to form amide bonds, which weaken the hydrophilic ability of E-conea and thus enhance the hydrophobicity of the coating.²⁶ On the other hand, the introduction of E-conea may slightly reduce the number of silanol groups on the surface of the coating, thereby enhancing the hydrophobic performance of the coating; hence, the contact angle increases. For the physically doped E-conea sample S4c, the possible reason is that the crosslinking agent KH550 added to the system provides additional NH_2 , which enhances the hydrophilicity of the coating,

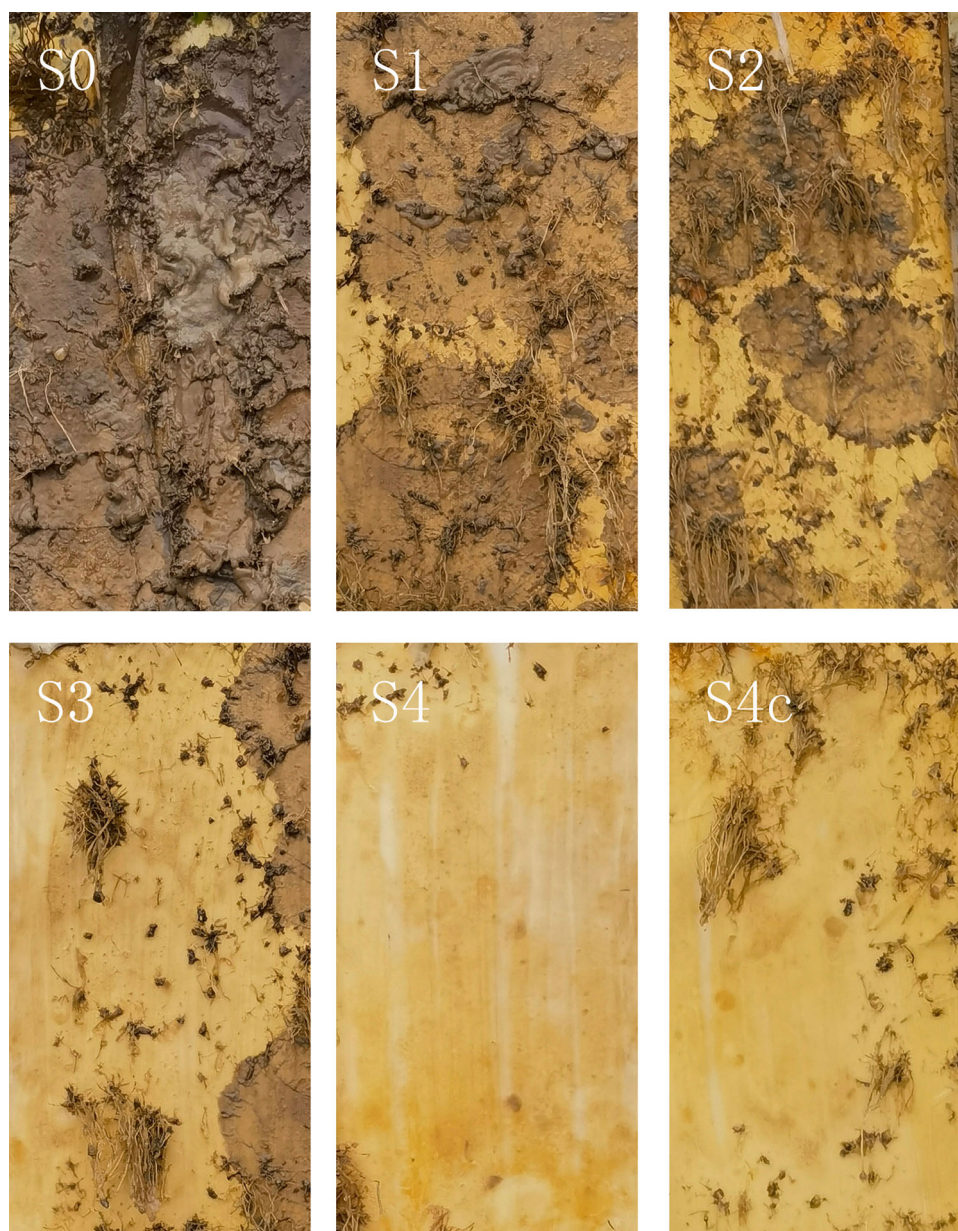


Fig. 7: Marine field test of coatings in Zhoushan Sea for 141 days to examine the antifouling mechanism

resulting in a lower contact angle than that of the chemically modified samples.

Surface roughness analysis

To further investigate the impact of morphology on antifouling performance, we conducted ultra-depth-of-field electron microscopy observations on the samples used, studying the surface roughness of both dry and wet film samples (Fig. 4). In the dry film state, the average surface roughness (S_a) of all samples was approximately between 10 and 12 μm , indicating that the introduction of 0.5–2.0% Econeal did not signifi-

cantly alter the roughness. After the coated samples were immersed in water for 7 days, the results varied. The surface roughness of the coating sample S0 without Econeal showed almost no significant change. The average surface roughness of the chemically grafted samples S1, S2, S3, and S4 decreased sequentially; for instance, the average surface roughness of S1, S2, S3, and S4 was 10.16 μm , 5.96 μm , 1.79 μm , and 1.36 μm , respectively. Notably, the physically blended sample S4c had an average roughness of 10.60 μm in the dry film state, which increased to 17.5 μm after being immersed in water for 7 days, a 65% increase. This was the highest average roughness among all samples and may be due to the hydrophilic Econeal

continuously seeping outward, forming erosion layer cavities on the coating surface, thereby increasing the surface roughness.²⁷ This is also consistent with the reports in the literature and further proves that the chemical grafting of Econeal is essential.²⁰

Release test of Econeal

The results presented in Table 2 reveal that the release rate of Econeal in the ASW obtained from a silicone coating with grafted Econeal (S4) was sixfold lower than the content obtained from a silicone coating with Econeal incorporated (S4c). These results confirm the chemical immobilization strategy effectiveness, which led to the biocide-release minimization in the silicone coating.²⁸ However, it should be also considered that due to the high instability of Econeal in water the released biocide content may be underestimated. Some biocide releasing extension can be also associated with the isocyanate group reaction with other coating formulation components or even with the presence of moisture during the coating process, which might lead to a lower grafting ability with the polymeric framework of the PDMS coating.¹⁹ Nevertheless, and since the releasing tests were performed at the same time and conditions for different strategies, the obtained chemical immobilization effectiveness cannot be disregarded.

Algae observation analysis

The laboratory antifouling performance of all samples was assessed against marine microalgae, with all samples being exposed to a microalgae suspension for 48 h (Fig. 5). The algae inhibition rate was calculated by ImageJ (Fig. 6). When the Econeal content was 0.5%, the algae inhibition capacity was only 14% compared to the control sample S0 (without Econeal), and there was no significant difference. As the Econeal content gradually increased, the algae inhibition capability significantly strengthened. Interestingly, when the Econeal addition reached 2%, both physically doped and chemically modified samples demonstrated excellent antialgae effects (about 91% antialgae rate). The release rate of Econeal from chemically modified samples is typically only one-sixth that of physically doped samples and can maintain long-lasting antifouling effects, which has been confirmed in previously published literature.²⁸

Antifouling performance testing

The antifouling performance of polymers was traditionally assessed through laboratory tests using marine microalgae. However, such conditions are quite different from those in the actual marine environment. Here, the marine antifouling ability of all samples was

examined through field tests (Fig. 7). The adhesion area and antifouling performance of the fouling organisms were calculated by ImageJ (Fig. 8). After being immersed in seawater for 141 days, the control panel without Econeal was seriously fouled by mud, algae, barnacles, and other fouling organisms, which can be considered to have a fouling rate of 100%.

From S1–S4, as the Econeal content increases, the antifouling performance gradually improves. When the Econeal content reached 2%, the biofouling resistance of S4 and S4c reached 92.1% and 75.9%, respectively. S4 exhibits the best antifouling performance, but its compatibility is the poorest among all samples, as evidenced by the gray-white streaks observed in the shallow sea panel photos. A possible explanation is that during film formation, the ethyl acetate content in the system progressively increases from S1 to S4, while xylene decreases correspondingly. This results in a continuous enhancement of the system's polarity, thereby reducing compatibility. The superior antifouling performance of S4c compared to S3 demonstrates that even without grafting Econeal, effective antifouling can still be achieved through chemical blending by maximizing the incorporation of antifouling agents. Furthermore, the shallow sea panel tests also demonstrate that the chemical bonding approach using coupling agents does not significantly alter the original antifouling performance of the antifouling agents.

Fig. 9 shows the antifouling mechanism of the tailored Econeal fouling-release coating applied on the substrate. The Econeal grafted onto the side chain acts as an effective antifouling biocidal group. The low elastic modulus of the silicone compound provides a smooth surface, which further reduces the attachment rate of fouling organisms. The combination of these synergistic effects leads to a high antifouling performance of prepared polymer coatings in seawater. In addition, this study further demonstrates that Econeal, when chemically grafted into the silicone system, can still maintain high-efficiency antifouling performance.

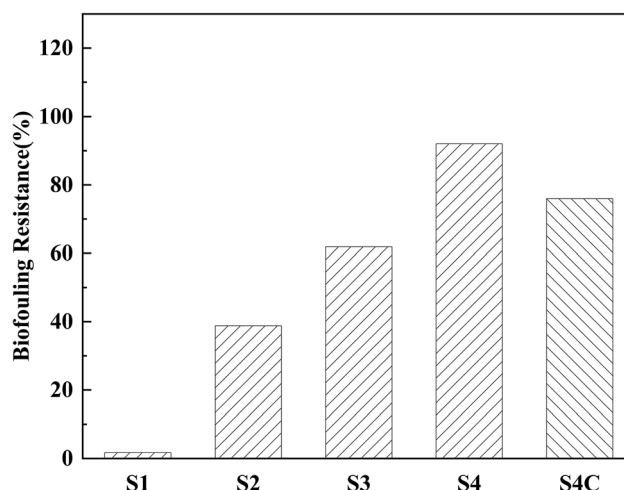


Fig. 8: Antifouling biological properties

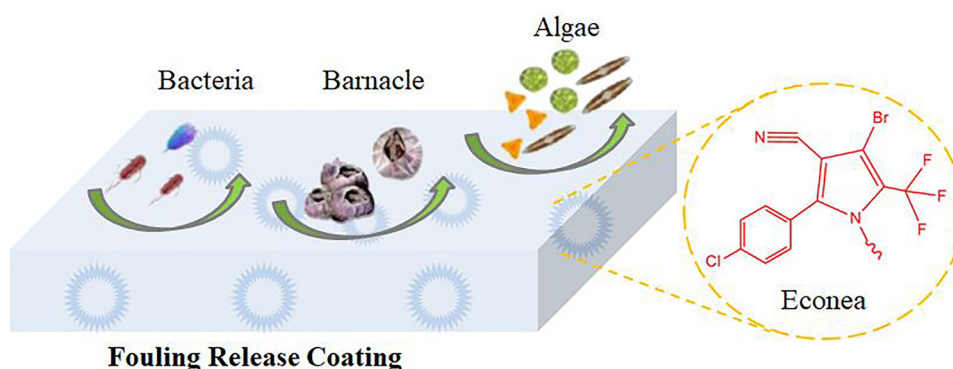


Fig. 9: Schematic of the antifouling mechanism of the coating

Conclusions

This study proposes a simple and convenient method for grafting an antifoulant onto a low surface coating by using a silane coupling agent to introduce Econeal into it. Compared to traditional antifoulant-releasing coatings, the seepage rate of the grafted antifoulant coating is only one-sixth of that of the traditional coatings, and it has maintained satisfactory antifouling performance over the 141 days of actual marine testing. This study also demonstrated that introducing Econeal into low-surface coatings via a coupling agent does not significantly alter its original antifouling properties. This finding not only resolves industry doubts but also provides a reference for the grafting technology of new antifouling agents.

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Conflict of interest The authors declare no conflict of interest.

References

- Kio, M, Klauda, J, "Advances in Emerging Hydrogel Fouling-Release Coatings for Marine Applications." *J. Coat. Technol. Res.*, **21** (3) 827–856 (2024)
- Hopkins, G, Davidson, I, Georgiades, E, Floerl, O, Morrissey, D, Cahill, P, "Managing Biofouling on Submerged Static Artificial Structures in the Marine Environment – Assessment of Current and Emerging Approaches." *Front. Marine Sci.*, **8** 759194 (2021)
- Qiu, HY, Feng, K, Gapeeva, A, Meurisch, K, Kaps, S, Li, X, Yu, LM, Mishra, YK, Adelung, R, Baum, M, "Functional Polymer Materials for Modern Marine Biofouling Control." *Prog. Polym. Sci.*, **127** 101516 (2022)
- Selim, MS, El-Safty, SA, Shenashen, MA, Higazy, SA, Elmarakbi, A, "Progress in Biomimetic Leverages for Marine Antifouling Using Nanocomposite Coatings." *J. Mater. Chem. B*, **8** (17) 3701–3732 (2020)
- Galli, G, Martinelli, E, "Amphiphilic Polymer Platforms: Surface Engineering of Films for Marine Antibiofouling." *Macromol. Rapid Commun.*, **38** (8) 1600704 (2017)
- Yang, WJ, Neoh, K-G, Kang, E-T, Teo, SLM, Rittschof, D, "Polymer Brush Coatings for Combating Marine Biofouling." *Prog. Polym. Sci.*, **39** (5) 1017–1042 (2014)
- Zhu, GY, Kang, FY, Tan, QY, Li, HL, Zhang, GG, Wu, M, Tan, Y, Liu, FQ, "Construction of Copper-Functionalized Two-Dimensional Zeolitic Imidazole Framework-7 (ZIF-7) Nanosheets for Enhanced Antibacterial and Antibiofilm Performance." *Prog. Org. Coat.*, **190** 108383 (2024)
- Zhou, AJ, Li, SC, Sand, W, Wang, XX, Geng, YJ, Ban, XZ, Jin, ZQ, Zhang, RY, "Self-Growing Composite Photocatalytic Materials for Surface Self-Renewal of Marine Antifouling Coatings." *Sep. Purif. Technol.*, **354** 129387 (2025)
- Tong, ZM, Gao, F, Chen, SF, Song, LN, Hu, JK, Hou, Y, Lu, JG, Leung, MKH, Zhan, XL, Zhang, QH, "Slippery Porous-Liquid-Infused Porous Surface (SPIPS) with On-Demand Responsive Switching Between 'Defensive' and 'Offensive' Antifouling Modes." *Adv. Mater.*, **36** (9) 2308972 (2024)
- Zhang, Y, Xie, Q, Pan, J, Ma, C, Zhang, G, "Strong-Adhesion and Nonfouling Self-Generating Zwitterionic Janus Hydrogel Paint." *Chem. Eng. J.*, **485** 149959 (2024)
- Hu, P, Xie, Q, Ma, C, Zhang, G, "Fouling Resistant Silicone Coating with Self-Healing Induced by Metal Coordination." *Chem. Eng. J.*, **406** 126870 (2021)
- Xie, Q, Zeng, H, Peng, Q, Bressy, C, Ma, C, Zhang, G, "Self-Stratifying Silicone Coating with Nonleaching Antifoulant for Marine Anti-Biofouling." *Adv. Mater. Interfaces*, **6** (13) 1900535 (2019)
- Sommer, S, Ekin, A, Webster, DC, Stafslie, SJ, Daniels, J, VanderWal, LJ, Thompson, SE, Callow, ME, Callow, JA, "A Preliminary Study on the Properties and Fouling-Release Performance of Siloxane-Polyurethane Coatings Prepared from Poly(Dimethylsiloxane) (PDMS) Macromers." *Biofouling*, **26** (8) 961–972 (2010)
- Hao, X, Zhang, X, Wang, W, Liu, W, Lu, G, Zheng, W, Gong, D, Zhao, H, "Controlled Release of 1,2-Benzisothiazolin-3-One-Based Anion Encapsulated in Mg Al Layered Double Hydroxides for Antibacterial and Antifouling Applications." *Prog. Org. Coat.*, **189** 108208 (2024)
- Liu, W, Lou, T, Bai, X, He, X, Yuan, C, "Experimental and Molecular Dynamics Simulation Study on Antifouling Performance of Antimicrobial Peptide-Modified Aluminum Alloy Surfaces." *Surf. Interface Anal.*, **56** (8) 532–543 (2024)
- Donnelly, B, Sammut, K, Tang, YH, "Materials Selection for Antifouling Systems in Marine Structures." *Molecules*, **27** (11) 3408 (2022)
- Detty, MR, Ciriminna, R, Bright, FV, Pagliaro, M, "Xerogel Coatings Produced by the Sol-Gel Process as Anti-Fouling,

- Fouling-Release Surfaces: From Lab Bench to Commercial Reality." *ChemNanoMat*, **1** (3) 148–154 (2015)
18. Kabir, R, "Circular Initiatives for the European Coating Industry." Master's Thesis, Tampere University of Applied Sciences (2024)
19. Ferreira, O, Rijo, P, Gomes, JF, Santos, R, Monteiro, S, Vilas-Boas, C, Correia-da-Silva, M, Almada, S, Alves, LG, Bordado, JC, Silva, ER, "Biofouling Inhibition with Grafted Econe Biocide: Toward a Nonreleasing Eco-Friendly Multiresistant Antifouling Coating." *ACS Sustain. Chem. Eng.*, **8** (1) 12–17 (2020)
20. Wang, HY, Wang, JH, Cui, M, Su, RX, Huang, RL, "The Preparation of Antifouling and Superhydrophobic Polyurethane Sponge by Grafting Econe to Diisocyanate Monomer for Durable Continuous Oil/Water Separation and Cleanup of Oil Spills." *Sep. Purif. Technol.*, **316** 123826 (2023)
21. Silva, ER, Tulcidas, AV, Ferreira, O, Bayon, R, Igartua, A, Mendoza, G, Mergulhao, FJM, Faria, SI, Gomes, LC, Carvalho, S, Bordado, JCM, "Assessment of the Environmental Compatibility and Antifouling Performance of an Innovative Biocidal and Foul-Release Multifunctional Marine Coating." *Environ. Res.*, **198** 111219 (2021)
22. Lejars, M, Margailan, A, Bressy, C, "Fouling Release Coatings: A Nontoxic Alternative to Biocidal Antifouling Coatings." *Chem. Rev.*, **112** (8) 4347–4390 (2012)
23. ASTM International, "Standard Practice for Preparation of Substitute Ocean Water." ASTM D1141-98; ASTM International, West Conshohocken, 2021.
24. Song, B, Zhang, E, Han, X, Zhu, H, Shi, Y, Cao, Z, "Engineering and Application Perspectives on Designing an Antimicrobial Surface." *ACS Appl. Mater. Interfaces*, **12** (19) 21330–21341 (2020)
25. Rodríguez-Fabià, S, Torstensen, J, Johansson, L, Syverud, K, "Hydrophobisation of Lignocellulosic Materials Part I: Physical Modification." *Cellulose*, **29** (10) 5375–5393 (2022)
26. Pinto, MMM, Palmeira, A, Fernandes, C, Resende, D, Sousa, E, Cidade, H, Tiritan, ME, Correia-da-Silva, M, Cravo, S, "From Natural Products to New Synthetic Small Molecules: A Journey Through the World of Xanthenes." *Molecules*, **26** (2) 431 (2021)
27. Chen, RZ, Xie, QY, Zeng, HH, Ma, CF, Zhang, GZ, "Non-Elastic Glassy Coating with Fouling Release and Resistance Abilities." *J. Mater. Chem. A*, **8** (1) 380–387 (2020)
28. Silva, ER, Ferreira, O, Ramalho, PA, Azevedo, NF, Bayon, R, Igartua, A, Bordado, JC, Calhorda, MJ, "Eco-friendly Non-Biocide-Release Coatings for Marine Biofouling Prevention." *Sci. Total Environ.*, **650** (Pt 2) 2499–2511 (2019)

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